WORKED SOLUTION

STEP I 1990, Question 6

Pure • Vectors and Parallelograms

STEP I 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q6

The key idea

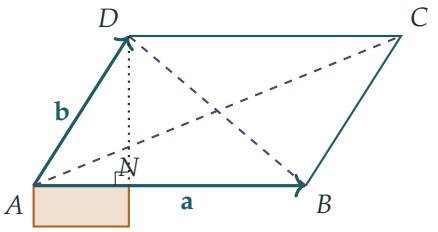
Put the origin at A and describe everything with the two side vectors $\mathbf{a} = \overrightarrow{AB}$ and $\mathbf{b} = \overrightarrow{AD}$. Then the diagonals are $\mathbf{a} + \mathbf{b}$ and $\mathbf{b} - \mathbf{a}$, and every squared length is a dot product of a vector with itself. Expanding those dot products does all the work. The only geometry needed is that $|\mathbf{u} \cdot \mathbf{v}|$ is the length of $\mathbf{u}$ times the length of the projection of $\mathbf{v}$ onto $\mathbf{u}$.

Solution

Take A as the origin, and let $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{b}$. Since $ABCD$ is a parallelogram, $\overrightarrow{BC} = \overrightarrow{AD} = \mathbf{b}$, so

$$\overrightarrow{AC} = \mathbf{a} + \mathbf{b}, \qquad \overrightarrow{BD} = \mathbf{b} - \mathbf{a}.$$

For any vector, $|\mathbf{v}|^2 = \mathbf{v} \cdot \mathbf{v}$, and the dot product is symmetric and distributes over addition. Write $a = |\mathbf{a}|$ and $b = |\mathbf{b}|$.



In the figure, N is the foot of the perpendicular from D to the line AB, so AN is the projection of AD on AB, with length $|\mathbf{a} \cdot \mathbf{b}|/a$.

Part (i)

Opposite sides are equal, so the left-hand side is $2a^2 + 2b^2$. For the diagonals,

$$AC^2 = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = a^2 + 2\,\mathbf{a}\cdot\mathbf{b} + b^2,$$
$$BD^2 = (\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a}) = a^2 - 2\,\mathbf{a}\cdot\mathbf{b} + b^2.$$

Adding, the cross terms cancel and $AC^2 + BD^2 = 2a^2 + 2b^2 = AB^2 + BC^2 + CD^2 + DA^2$, as required. (This is often called the parallelogram law.)

Part (ii)

Subtracting the same two lines instead,

$$AC^2 - BD^2 = 4\,\mathbf{a}\cdot\mathbf{b}, \qquad \text{so} \qquad \left|AC^2 - BD^2\right| = 4\,|\mathbf{a}\cdot\mathbf{b}|.$$

Now interpret $|\mathbf{a}\cdot\mathbf{b}|$. If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then $|\mathbf{a}\cdot\mathbf{b}| = a\cdot b|\cos\theta|$, and $b|\cos\theta|$ is exactly the length AN of the projection of AD on AB. So $|\mathbf{a}\cdot\mathbf{b}|$ is the area of the rectangle (shaded in the figure) with sides AB and the projection of AD on AB.

Ask yourself The result claims this works for *any* side and *either* adjacent side. Why don't we need eight separate calculations?

Every side of the parallelogram is a vector $\pm\mathbf{a}$ or $\pm\mathbf{b}$, and the two sides adjacent to it are the other vector, up to sign. For example, side BC is $\mathbf{b}$, and its neighbours AB and CD are $\mathbf{a}$ and $-\mathbf{a}$. Changing signs doesn't change the lengths of the vectors or of the projection, since the projection has length $|\mathbf{u}\cdot\mathbf{v}|/|\mathbf{u}|$. So every choice of side and adjacent side gives the same rectangle area $|\mathbf{a}\cdot\mathbf{b}|$: either $a\times$ (projection of $\mathbf{b}$ on $\mathbf{a}$) or $b\times$ (projection of $\mathbf{a}$ on $\mathbf{b}$), and both equal $|\mathbf{a}\cdot\mathbf{b}|$. Hence, in every case,

$$\left|AC^2 - BD^2\right| = 4\times(\text{area of that rectangle}). \qquad \blacksquare$$

The final part

Part (ii) came from the dot product of the two *sides*. Swap the roles of sides and diagonals and take the dot product of the two *diagonals*:

$$\overrightarrow{AC}\cdot\overrightarrow{BD} = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{b}-\mathbf{a}) = b^2 - a^2,$$

because the cross terms $\mathbf{a}\cdot\mathbf{b}$ and $-\mathbf{b}\cdot\mathbf{a}$ cancel. Therefore

$$\left|AB^2 - AD^2\right| = |\overrightarrow{AC}\cdot\overrightarrow{BD}|.$$

Arguing as in part (ii), this gives the following.

Result. $\left|AB^2 - AD^2\right|$ equals the area of the rectangle whose sides are either diagonal and the projection of the other diagonal on it.

Proof. If φ is the angle between the diagonals, $|\overrightarrow{AC} \cdot \overrightarrow{BD}| = AC \cdot BD\,|\cos\varphi|$. Here $BD\,|\cos\varphi|$ is the length of the projection of BD on AC, and $AC\,|\cos\varphi|$ is the length of the projection of AC on BD. So the area is the same whichever diagonal we start from, and it equals $\left|AB^2 - AD^2\right|$ by the display above. ■

To make it look even more like (ii), let O be the point where the diagonals bisect each other. Then $AC = 2\,OA$ and the projection of BD on AC is twice the projection of OB on OA. So $\left|AB^2 - AD^2\right|$ is also 4 times the area of the rectangle whose sides are OA and the projection of OB on OA: half-diagonals now play the part of the sides.

Two pleasant corollaries fall out. The diagonals are equal exactly when $\mathbf{a} \cdot \mathbf{b} = 0$, so exactly when the parallelogram is a rectangle. They are perpendicular exactly when $a = b$, so exactly when it is a rhombus.

Checks

Take $A = (0,0)$, $B = (4,0)$, $D = (1,2)$, so $C = (5,2)$. Then $AC^2 = 29$ and $BD^2 = 13$, and $2(16+5) = 42 = 29 + 13$, which checks (i). For (ii), $AC^2 - BD^2 = 16$, while the projection of AD on AB has length 1, so the rectangle has area $4 \times 1 = 4$, and $16 = 4 \times 4$. For the final part, $AB^2 - AD^2 = 16 - 5 = 11$, and $\overrightarrow{AC} \cdot \overrightarrow{BD} = (5,2) \cdot (-3,2) = -11$. We also tested all three results, for every choice of side and adjacent side, on 10^5 random parallelograms; the largest discrepancy was about 10^{-13}, which is just rounding error.

Common slips

- Writing $\overrightarrow{BD} = \mathbf{a} - \mathbf{b}$ or $\mathbf{a} + \mathbf{b}$. The sign doesn't matter for BD^2, but getting the direction right avoids confusion later.
- Forgetting the modulus: $\mathbf{a} \cdot \mathbf{b}$ is negative when the angle at A is obtuse, but an area can't be, which is why the question has $\left|AC^2 - BD^2\right|$.
- Giving a factor of 4 in the final part while using whole diagonals. With whole diagonals the factor is 1; the factor 4 appears only with half-diagonals.

Beyond STEP

The identity $|\mathbf{a} + \mathbf{b}|^2 - |\mathbf{a} - \mathbf{b}|^2 = 4\,\mathbf{a} \cdot \mathbf{b}$ is the *polarisation identity*: it shows that the dot product can be recovered from lengths alone. Together with the parallelogram law of part (i), it is a key fact about inner product spaces. A norm comes from an inner product exactly when it satisfies the parallelogram law, which is a theorem of Jordan and von Neumann. Strang's *Introduction to Linear Algebra* is a good place to see dot products, projections and orthogonality developed in any number of dimensions.

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